Progressive exercises

E1 · Practice. A code is a sequence of four digits chosen from $0, 1, \ldots, 9$. Repetitions and the digit 0 in first position are allowed. How many codes contain at least one 0?

E2 · Practice. Ten athletes take part in a race with no ties. How many different podiums, with a first, a second and a third place, can be obtained?

E3 · Practice. From seven distinct people, a group of three people is formed containing one person designated in advance. How many groups can be formed?

E4 · Practice. On a grid, we start from $(0;0)$ and reach $(4;3)$ using exactly four moves of one unit to the right and three upwards. How many different paths can be followed?

E5 · Practice. A class has eight girls and seven boys. How many groups of four students containing exactly two girls can be formed?

Textbook or past-paper work

Barbazo, p. 294, exercise 37. At home after S35: do both questions. State whether order matters and whether a card may be chosen several times before using a binomial coefficient.